

University of Dhaka
Institute of Leather Engineering and Technology
4th Year 2nd Semester B.Sc. in Leather Engineering Examination, 2024
Course Name: **Total Quality Management (TQM) for Leather and Leather Products**
Course No.: BUS-401 (3 Credit Points)

Time: 3 Hours.

Full Marks: 60

Answer any five questions of the following. The figure in the right-hand margin indicates full marks. All parts of a question must be answered sequentially.

1. Suppose, X Products Ltd. is introducing leather goods to the regional market. The company's monthly goal is to sell 30,000 quality leather bags. As a member of the senior management team of the company, your main focus is to draw clients for new business, hence answer the followings:
- (a) What should be included in a Quality policy statement? 3
 - (b) How would you implement Total Quality Management within an organization? 4
 - (c) Narrate some guidelines to build a customer-focused business. 5
2. An accessories supplier to shoe manufacture has little influence on profitability, but its main leather supplier has a huge impact. Any risk of the leather manufacturer is a major concern to the shoe manufacturer. As you are recruited to boost buyer-supplier networks, response the following:
- (a) What is Customer Orientation? State the importance of Customer orientation. 1+2
 - (b) Discuss different types of supplier and show which one is the best for business continuity, operational excellence, and profitability. 3
 - (c) What are the challenges associated with supplier relationship management? 3
 - (d) Consider, A, B and C are the three reputed suppliers, which play a vital role in your company have the following historical data: 3

Factors	Weight	A	B	C
		Score	Score	Score
Material Quality	30%	80	70	90
Material Delivery	25%	60	75	60
Price	20%	50	60	50
Out Look	10%	70	60	50
In Time Delivery	15%	50	40	30

- (i) Find out the best supplier through weighted point method
3. (a) What is meant by customer retention? Write down excellent customer retention strategies to build trust and enhance customer loyalty. 1+4
- (b) Narrate popular eight strategies to retain customers. 3
- (c) How could you calculate the following customer retention matrix for the business? 4
- Customer retention rate ii) Customer churn rate iii) Customer life time value iv) Customer satisfaction score
4. (a) What is sigma (6σ)? Give the feature of six sigma. 3
- (b) Describe the six sigma methodology DMAIC in details. 4
- (c) Write shortly the Benchmarking process to improve the situation. 5
5. (a) Distinguish between quality control and quality assurance? 2
- (b) Discuss the central limit theorem. 3

- (c) The weights of boxes of oat flakes within a large production lot are sampled each hour. Managers want to set control limits that include 99.73% of the sample means. For hour 1, randomly selected 9 boxes weight are 17 oz, 13 oz, 16 oz, 18 oz, 17 oz, 16 oz, 15 oz, 17 oz and 16 oz. Total 12 hourly samples are-

Hour	Ave. wt. of 9 boxes	Hour	Ave. wt. of 9 boxes	Hour	Ave. wt. of 9 boxes
1	16.1	5	16.5	9	16.3
2	16.8	6	16.4	10	14.8
3	15.5	7	15.2	11	14.2
4	16.5	8	16.4	12	17.3

Here, standard deviation is 1 ounce. Is the process being suitable or not? Determine.

- (d) How do you prepare a control charts? 3
6. (a) Define design of experiments (DOE)? How do you prepare DOE for an industry? 3
- (b) Distinguish between process capability ratio (C_p) and process capability index (C_{pk}). In a GE insurance claims process, $\bar{x} = 210$ minutes, and $\sigma = .516$ minutes. The design specification to meet customer expectations is 210 ± 3 minutes. Compute the process capability ratio. 4
- (c) What is operating characteristics (OC) curve? Illustrate a figure and define the following term- 5
- Acceptable quality level
 - Lot tolerance percentage defective
 - Producer's risk
 - Consumer's risk
7. (a) Define quality circles. Write down the feature of quality circles. 4
- (b) What is quality function deployment (QFD)? Discuss the main phases of QFD. 5
- (c) What is house of quality (HOQ)? Which components is essential to facilitate the QFD? 3

University of Dhaka
Institute of Leather Engineering and Technology
4th Year 2nd Semester B.Sc. in Leather Engineering Examination, 2024
Course Name: **Tannery Wastewater Management**
Course No.: LE-409 (3 Credit Points)

Time: 3 Hours.

Full Marks: 60

Answer any five questions of the following. The figure in the right-hand margin indicates full marks. All parts of a question must be answered sequentially.

1. (a) Define the following terms: 4
Pollution, contaminant, eutrophication, and bioaccumulation.
(b) Illustrate primary and secondary types of contaminants for water pollution. 3
(c) What are the principal industries and sectors attributed to water pollution? Mention the pollutants of each sector and their impacts on environment and human health. 5
2. (a) What is water pollution? Why does water polluted easily? 2.5
(b) Point out the measures to be taken for prevention of water pollution. 3.5
(c) Discuss the hazardous effects and impacts of pollution on land, ground water, surface water and aquatic life. 6
3. (a) Illustrate the effluents generated from beam house, tan yard and finishing processes, and their nature. 5
(b) Which are the most toxic ingredients in tannery wastewater? Explain their impact on ecosystem. 3
(c) Through lights on characterization and assessment of critical environmental parameters. 4
4. (a) Describe the major processing steps in tanneries that generate liquid wastes. 3
(b) Briefly discuss the importance of wastewater treatment in leather processing. 4
(c) Define CBOD and NBOD. Why does NBOD typically appear later in the standard 5-day BOD test? 5
5. (a) Differentiate between MLSS and MLVSS. 2
(b) What are the objectives of primary treatment of wastewater? 2
(c) Why is jar testing necessary? If jar testing indicates a requirement of 50 mg/L alum, calculate the total mass of alum needed for 1 MLD of wastewater. 2
(d) Explain the mechanism of coagulation in wastewater treatment. 6
6. (a) What is the Activated Sludge Process (ASP)? Explain its major components. 2
(b) Define the following terms in Activated Sludge Process: 6
i) Sludge Volume Index (SVI)
ii) Sludge age
iii) F/M ratio
(c) Solve the given problem: 4
A completely mixed activated sludge process is used to treat a wastewater flow of 1 MLD having a BOD₅ of 200 mg/L. The biomass concentration in the aeration tank is 3000 mg/L and the concentration of net biomass leaving the system per day is 50 mg/L. The volume of aeration tank is 200m³.
i) Determine the Hydraulic Retention Time (HRT) of the wastewater in the aeration tank.
ii) Determine the average time for which biomass stays in the system.
7. (a) Narrate the current status of the leather and leather products industry in Bangladesh. 4
(b) State the biological wastewater treatment and its principles. 4
(c) Mention the advantages and disadvantages of a trickling filter. 4

Worker	Day 1	Day 2	Day 3	Day 4	Day 5
A	180	190	200	210	220
B	150	160	170	180	190
C	220	230	210	205	215

University of Dhaka
Institute of Leather Engineering and Technology
4th Year 2nd Semester B.Sc. in Leather Engineering Examination, 2024
Course Name: **Cost and Management Accounting**
Course No.: BUS-402 (3 Credit Points)

Time: 3 Hours.

Full Marks: 60

Answer any five questions of the following. The figure in the right-hand margin indicates full marks. All parts of a question must be answered sequentially.

1. (a) Define Cost Accounting and discuss its scope.

4

(b) Fine Hide Leather Company purchases large, uniquely identifiable batches of cowhides (measured in square meters) for cutting and processing. The company is evaluating alternative material costing methods for strategic planning. The transactions for the month of November are detailed below:

4

Date	Transaction	Batch No.	Units sq meter	Cost per sq meter (Tk)	Total Cost (Tk)
Nov. 1	Beginning Inventory	L-500	800	25.00	20,000
Nov. 7	Purchase	L-501	1,200	28.00	33,600
Nov. 15	Issue to Job P10	1,500	?	?	?
Nov. 20	Purchase	L-502	1,000	30.00	30,000
Nov. 25	Issue to Job Q25	500	?	?	?
Nov. 28	Purchase	L-503	700	32.00	22,400

The production supervisor's reports show the material usage for the jobs:

- The issue on November 15 for Job P10 specifically used:
 - 800 sq meters from the beginning inventory batch L-500
 - 700 sq meters from the November 7 purchase batch L-501
- The issue on November 25 for Job Q25 specifically used:
 - 500 sq meters from the remaining November 7 purchase batch L-501

On November 25, the replacement cost (market price) to purchase a sq meter of similar leather hide was Tk. 31.50 per sq meter. For the Nov. 15 issue, assume the replacement cost on that date was Tk. 29.00 per sq meter.

Required: Calculate the total direct material cost charged to Job P10 and Job Q25, and the value of the ending inventory on November 30 using the following two methods: a) Specific Identification Method, b) Current Cost (Replacement Cost) Method.

(c) A company sets the standard output of a worker at 200 units per day. Taylor's Differential Piece Rate System is applied with the following rates:

4

Lower piece rate: \$2.00 per unit (if output < 200 units)

Higher piece rate: \$3.00 per unit (if output ≥ 200 units)

During one week (5 working days), three workers produce the following outputs:

- Calculate the daily wages of each worker using Taylor's Differential Piece Rate System.
- Find the total weekly earnings of each worker.
- Compare the average earnings per day of Worker B and Worker C.

2. (a) Compare and contrast Cost Accounting, Financial Accounting, and Management Accounting. Provide relevant examples.

4

- (b) Chem Clean's maintenance costs are a mixed cost. Here is the data for the past six months, linking the volume of solvent produced to the total maintenance cost:

Month	Volume of Solvent Produced (Liters)	Total Maintenance Cost (Taka)
January	1,500	12,500
February	2,800	18,200
March	1,000	10,000
April	3,500	21,500
May	2,100	14,800
June	3,000	19,000

Use the data set above to calculate the fixed and variable components of the maintenance cost using both the High-Low Method and the Least Squares Regression Method.

- (c) A company manufactures filing cabinets. The company applies all manufacturing overhead to products based on direct labor cost. Selling and administrative expenses are treated as period costs. The company's total budgeted costs and activity for the year are as follows:

- Total Manufacturing Overhead: \$500,000
- Total Selling and Administrative Overhead: \$300,000
- Total units produced: 1,000 units
- Direct labor cost per unit: \$90

The company received an order from a customer, OfficeMart, with the following details:

- Total number of filing cabinets ordered: 80 units
- Selling price per unit: \$795
- Direct materials per unit: \$185
- Direct labor per unit: \$90

Requirement: Using the absorption costing method, calculate the gross margin for the Office Mart order.

3. (a) Discuss the characteristics of an ideal cost accounting system.

- (b) A company, Precision Tools Co., manufactures specialized wrench sets. The company applies all manufacturing overhead to products based on direct labor cost. Selling and administrative expenses are treated as period costs. The company's total budgeted costs and activity for the year are as follows:

- Total Manufacturing Overhead: \$600,000
- Total Selling and Administrative Overhead: \$250,000
- Total units produced: 2,000 units
- Direct labor cost per unit: \$75

Precision Tools Co. received a custom order from a retailer, The Hardware Depot, with the following details:

- Total number of wrench sets ordered: 150 units
- Selling price per unit: \$550
- Direct materials per unit: \$120
- Direct labor per unit: \$75

Requirement: Using the absorption costing method, calculate the gross margin for The Hardware Depot order.

- (c) Luxguard Home Paint Company produces exterior latex paint, which it sells in one-gallon containers. The company has two processing departments—Base Fab and Finishing. White paint, which is used as a base for all the company's paints, is mixed from raw ingredients in the Base Fab Department. Pigments are then added to the basic white paint, the pigmented paint is squirted under pressure into one-gallon containers, and the containers are labeled and packed for shipping in the Finishing Department. Information relating to the company's operations for April follows:

- Issued raw materials for use in production: Base Fab Department, \$851,000; and Finishing Department, \$629,000.
- Incurred direct labor costs: Base Fab Department, \$330,000; and Finishing Department, \$270,000.

- iii. Applied manufacturing overhead cost: Base Fab Department, \$665,000; and Finishing Department, \$405,000.
- iv. Transferred basic white paint from the Base Fab Department to the Finishing Department, \$1,850,000.
- v. Transferred paint that had been prepared for shipping from the Finishing Department to Finished Goods, \$3,200,000.

Requirement:

1. Prepare journal entries to record items (i) through (v) above.

Post the journal entries from (1) above to T-accounts.

4. (a) Compare and contrast Job Order Costing and Process Order Costing. 2
- (b) What is cost object? How can we classify costs for assigning costs to cost objects? Provide relevant examples. 4
- (c) What is cost driver and relevant range? Discuss the cost classifications for predicting cost behavior. 3
- (d) Discuss the usefulness of the standard costing and variance analysis in performance measurement and evaluation. Consider a company's standard material cost is 2 kg at \$10/kg. Actual production used is 420 kg at \$9.50/kg for an output that should have used 400 kg. Measure the company performance using variance analysis. 3

5. (a) i. A company, AgriPure Processing, processes raw vegetable oil into two joint products: Cooking Oil (CO) and Industrial Lubricant (IL). The company uses the net realizable value (NRV) method to allocate joint costs. The processing of a single batch yielded the following results for October: 4

Product	Units Produced (Liters)	Selling Price at Split-Off	Separable Processing Costs per Unit	Final Selling Price per Unit
CO	10,000	\$4.00	\$0.50	\$5.50
IL	5,000	\$2.50	\$0.75	\$4.00

Total Joint Manufacturing Costs for the batch were \$35,000. Using the Net Realizable Value (NRV) method, calculate the total joint cost allocated to Cooking Oil (CO).

- ii. The standard cost for the Cooking Oil (CO) produced was based on the following:

- Standard Quantity (SQ): 1.25 kg of raw oil per liter of CO
- Standard Price (SP): \$0.80 per kg of raw oil

The actual results for the 10,000 liters of CO produced were:

- Actual Quantity (AQ): 13,000 kg of raw oil purchased and used
- Actual Cost (AC): \$10,140 (Total cost for 13,000 kg)

Calculate the following for the raw materials used in Cooking Oil (CO): a. The Direct Materials Price Variance. b. The Direct Materials Quantity (Usage) Variance.

- (b) Three grams of musk oil are required for each bottle of Mink Caress, a very popular perfume made by a small company in western Siberia. The cost of the musk oil is \$1.50 per gram. Budgeted production of Mink Caress is given below by quarters for Year 2 and for the first quarter of Year 3: 4

	Year 2				Year 3
	First	Second	Third	Fourth	First
Budgeted production, in bottles	60,000	90,000	150,000	100,000	70,000

Musk oil has become so popular as a perfume ingredient that it has become necessary to carry large inventories as a precaution against stock-outs. For this reason, the inventory of musk oil at the end of a quarter must be equal to 20% of the following quarter's production needs. Some 36,000 grams of musk oil will be on hand to start the first quarter of Year 2.

- (c) The marketing department of Jessi Corporation has submitted the following sales forecast for the upcoming fiscal year (all sales are on account): 4

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
Budgeted unit sales	11,000	12,000	14,000	13,000

The selling price of the company's product is \$18.00 per unit. Management expects to collect 65% of sales in the quarter in which the sales are made, 30% in the following quarter, and 5% of sales are expected to be uncollectible. The beginning balance of accounts receivable, all of which is expected to be collected in the first quarter, is \$70,200. The company expects to start the first quarter with 1,650 units in finished goods inventory. Management desires an ending finished goods inventory in each quarter equal to 15% of the next quarter's budgeted sales. The desired ending finished goods inventory for the fourth quarter is 1,850 units.

Requirement:

- i. Prepare the company's sales budget and schedule of expected cash collections.

Prepare the company's production budget for the upcoming fiscal year.

6. (a) Define Management Accounting. Compare and contrast between Management Accounting and Cost Accounting. 3
- (b) Flight Café is a company that prepares in-flight meals for airlines in its kitchen located next to the local airport. The company's planning budget for July appears below: 5

Flight Café Planning Budget For the Month Ended July 31	
Budgeted meals (q)	18,000
Revenue (\$4.50q)	<u>\$81,000</u>
Expenses:	
Raw materials (\$2.40q)	43,200
Wages and salaries (\$5,200 + \$0.30q)	10,600
Utilities (\$2,400 + \$0.05q)	3,300
Facility rent (\$4,300)	4,300
Insurance (\$2,300)	2,300
Miscellaneous (\$680 + \$0.10q)	2,480
Total expense	<u>66,180</u>
Net operating income	<u>\$14,820</u>

In July, 17,800 meals were actually served. The company's flexible budget for this level of activity appears below:

Flight Café Flexible Budget For the Month Ended July 31	
Budgeted meals (q)	17,800
Revenue (\$4.50q)	<u>\$80,100</u>
Expenses:	
Raw materials (\$2.40q)	42,720
Wages and salaries (\$5,200 + \$0.30q)	10,540
Utilities (\$2,400 + \$0.05q)	3,290
Facility rent (\$4,300)	4,300
Insurance (\$2,300)	2,300
Miscellaneous (\$680 + \$0.10q)	2,460
Total expense	<u>65,610</u>
Net operating income	<u>\$14,490</u>

Requirement:

1. Prepare a report showing the company's activity variances for July.
 2. Which of the activity variances should be of concern to management? Explain.
- (c) Discuss the risks associated with foreign exchange exposure and the methods adopted to mitigate those risks. 4

7. (a)
(b)

Why is knowledge of cost behavior needed in product costing?

2
10

i. Last month when Holiday Creations, Inc., sold 50,000 units, total sales were \$200,000, total variable expenses were \$120,000, and fixed expenses were \$65,000.

Required:

- What is the company's contribution margin (CM) ratio?
- What is the estimated change in the company's net operating income if it can increase total sales by \$1,000?

ii. Mauro Products distributes a single product, a woven basket whose selling price is \$15 per unit and whose variable expense is \$12 per unit. The company's monthly fixed expense is \$4,200.

Required:

- Calculate the company's break-even point in unit sales.
- Calculate the company's break-even point in dollar sales.
- If the company's fixed expenses increase by \$600, what would become the new break-even point in unit sales? In dollar sales?

iii. Engberg Company installs lawn sod in home yards. The company's most recent monthly contribution format income statement follows:

Particulars	Amount	Percent of Sales
Sales	\$80,000	100%
Variable expenses	32,000	40%
Contribution margin	48,000	60%
Fixed expenses	38,000	
Net operating income	\$10,000	

Required:

- What is the company's degree of operating leverage?
- Using the degree of operating leverage, estimate the impact on net operating income of a 5% increase in sales.
- Verify your estimate from part (b) above by constructing a new contribution format income statement for the company assuming a 5% increase in sales.

Institute of Leather Engineering and Technology

University of Dhaka

4th Year, 2nd Semester B. Sc. in Leather Engineering

1st In-course Assessment Exam-2024

Subject: Total Quality Management for Leather and Leather Products (BUS-401)

Time: 1 hour

Full Marks: 30

1. Define quality. What are the determinants of service quality. 4
2. How do you improve productivity through quality? 4
3. Discuss the types of quality. 6
4. Explain Deming's 14 points for management. 5
5. How do you implement TQM in leather industry? 5
6. What is cost of quality (COQ)? Classify COQ with examples. 6

Institute of Leather Engineering and Technology, University of Dhaka
4th Year 2nd Semester BSc in Leather Engineering
2nd Incourse Examination-2024

Course Name: Tannery Wastewater Management; Course No.: LE-409 (3 Credits)

Total marks: 30

Time: 1 Hours

(Sequence must be maintained while answering the questions)

1. Define the following terms:
Pollution, contaminant, water pollution, eutrophication, and bioaccumulation. 05
2. Discuss different sources of water pollution. 04
3. Illustrate primary and secondary types of contaminants. 05
4. Illuminate various types of ~~source~~ of pollution. 07
5. Through lights on effects and impacts of water pollution on ecology and environment, i.e. land, ground water, surface water, aquatic life and sea. 05
6. Point out the measures to be taken for prevention of water pollution.

University of Dhaka

Institute of Leather Engineering and Technology

4th year 2nd semester B.Sc. in Leather Engineering, 1st in-course

Course Title: Tannery Wastewater Management Course code: LE- 409

Time: 1 hour

Marks: 30

1. What are the major water pollutants and solid pollutants generated during different stages of leather processing? 3
2. What are the three main categories of tannery wastewater? Why should liming and tanning wastewater streams not be mixed? 2+1
3. List the typical stages of treatment in an ETP and explain the purpose of each stage. 5
4. Determine the ultimate BOD of the wastewater sample which was subjected to BOD test in which 6 ml of wastewater containing 1.5 mg/L of DO was mixed with 294 ml of water containing 8.6 mg/L of DO. After incubation at 20 °C for 5 days, the DO of the mix was 5.4 mg/L. The BOD rate constant to the base e, $K_{20^\circ C} = 0.25 \text{ day}^{-1}$ 2/4.3 3
5. What are the chemical coagulation and mechanical flocculation processes? How do coagulants work in the coagulation process? 4
6. Solve the given problems: 3
 - i) Suppose, in a ETP the wastewater flow rate is 150 L/s. In the jar test, the optimum coagulation occurs when 1 liter of water is dosed with 15 mL of 20g/L alum solution, 2.5 ml of 10 g/L suspension of lime and 0.45 mg/l of polyelectrolyte. Determine the daily requirement of alum, lime and polyelectrolyte for that ETP.
 - ii) If a WWT plant treating 10 MLD of water requires 20 mg/l of alum dosages, then determine the following:
 - a) How much floc will be generated per day in the entire system?
 - b) How much CO₂ gas will be released per day from the plant.Here, 1 mg/l of alum generates 0.23 mg/l of floc and 0.396 mg/l of CO₂
8. Define the following terms: i) UBOD ii) CBOD iii) NBOD 6

University of Dhaka
Institute of Leather Engineering and Technology
4th Year, 2nd Semester B.Sc. in Leather Engineering (LE)
2nd In-course Assessment Exam-2024

Subject- Total Quality Management (TQM) for leather and leather products
Course Code (BUS-401)

Time: 01 Hour

Total Marks-30

(Answer any Six of the followings)

✓ 01. Assume, ABC products Ltd. is launching leather goods for local market. Target of the company is to sell 10,000 fancy leather bags per months. Being a top management 05
personnel develop the following statement for the new company to attract customer:

- a) Vision statement
- b) Mission statement and
- c) Quality policy statement

02. Show how to implement Total Quality Management for the various quality management 05
tools work together to achieve control throughout the process chain within an organization.

03. You are well-positioned for long-term success and growth through emphasis on realizing 05
and meeting consumer needs. Set some strong importance for developing a customer-
focused mindset in the company.

04. Top management and frontline staff must carefully plan for customer orientation in 05
business. To start prioritizing consumers, narrate the implementation steps of customer
orientation.

05. Let, a waist belt manufacturer's accessories supplier has little influence on profitability, 05
but its main leather supplier has a huge impact, making it a key strategic partner. Any risk
of the leather manufacturer is a major concern to the belt manufacturer.

- a) Discuss different types of supplier and show which one is the best for business
continuity, operational excellence and profitability.

06. A successful Supplier relationship management (SRM) discipline seeks not just cost 05
savings, but to maximize the value of suppliers to gain a competitive advantage and
improved brand existence.

- a) What do you mean by SRM?
- b) What could be the potential benefits of SRM?

07. What are the major challenges associated with supplier relationship management? 05

08. Suppliers are more than just vendors—they are key partners in your business ecosystem. 05
They affect in setting criteria for supplier selection and evaluation.

- a) Which criteria should procurement manager consider for supplier selection and
evaluation?

University of Dhaka
Institute of Leather Engineering and Technology
Leather Engineering Department
BSc in Leather Engineering 39th Batch 4th Year 2nd Semester 2nd Mid-term Examination –
2025

BUS 402: Cost and Management Accounting

Time: 1 Hour

Marks: 15

Answer all of the following questions:

Q-1. What is cost accounting? Compare and contrast among Cost accounting, Financial Accounting and Management Accounting. (4)

Q-3. What is job costing? We will consider a single month's activity at Ruger Corporation, a producer of gold and silver commemorative medallions. Ruger Corporation has two jobs in process during April, the first month of its fiscal year. Job A, a special minting of 1,000 gold medallions commemorating the invention of motion pictures, was started during March. By the end of March, \$30,000 in manufacturing costs had been recorded for the job. Job B, an order for 10,000 silver medallions commemorating the fall of the Berlin Wall, was started in April.

- i. On April 1, Ruger Corporation had \$7,000 in raw materials on hand. During the month, the company purchased on account an additional \$60,000 in raw materials.
- ii. During April, \$52,000 in raw materials were requisitioned from the storeroom for use in production. These raw materials included \$50,000 of direct and \$2,000 of indirect materials. (Where, \$28,000 of the \$50,000 in direct materials is charged to Job A's cost sheet and the remaining \$22,000 is charged to Job B's cost sheet.)
- iii. In April, the employee time tickets included \$60,000 recorded for direct labor and \$15,000 for indirect labor. (During April, \$40,000 of direct-labor cost was charged to Job A and the remaining \$20,000 was charged to Job B.)
- iv. Ruger Corporation incurred the following general factory costs during April- Utilities \$ 21,000, Rent on Factory Equipment \$ 16,000, and miscellaneous factory overhead costs \$ 3,000. It sums up to \$40,000 costs in total.
- v. In addition, assume that during April, Ruger Corporation recognized \$13,000 in accrued property taxes and that \$7,000 in prepaid insurance expired on factory buildings and equipment.
- vi. Finally, assume that the company recognized \$18,000 in depreciation on factory equipment during April.
- vii. Assume that Ruger Corporation's predetermined overhead rate is \$6 per machine-hour. Also assume that during April, 10,000 machine-hours were worked on Job A and 5,000 machine-hours were worked on Job B (a total of 15,000 machine-hours). Thus, \$90,000 in overhead cost (\$6 per machine-hour * 15,000 machine-hours = \$90,000) would be applied to Work in Process.
- viii. Ruger Corporation incurred \$30,000 in selling and administrative salary costs during April.
- ix. Depreciation on office equipment during April was \$7,000.
- x. Finally, assume that advertising was \$42,000 and that other selling and administrative expenses in April totaled \$8,000.
- xi. For Ruger Corporation, we will assume 750 of the 1,000 gold medallions in Job A were shipped to customers by the end of the month for total sales revenue of \$225,000. Because 1,000 units were produced and the total cost of the job from the job cost sheet was \$158,000, the unit product cost was \$158.

Requirement: Prepare journal entries to record the preceding transactions. Post the entries in (1) above to T-accounts. (11)

Required:

Calculate the total direct material cost charged to Job A101 and Job B202, and the value of the ending inventory on October 31st using the following two methods:

- a) Specific Identification Method
- b) Current Cost (Replacement Cost) Method (Use the replacement cost given for the issue on Oct. 25. For the Oct. 12 issue, assume the replacement cost on that date was £56.50 per kg).

University of Dhaka
Institute of Leather Engineering and Technology
Leather Engineering Department
BSc in Leather Engineering 4th Year 2nd Semester 1st Mid-term Examination – 2025
BUS 402: Cost and Management Accounting

Time: 1 Hour

Marks: 3+3+3+6

Answer all of the following questions:

1. What is Cost Driver? Discuss the cost classification for a manufacturing company with examples.
2. Compare and contrast among Halsey Plan, Rowan Plan, and Taylor's Differential Piece Rate System.
3. Using the information given in the following table, estimate the cost behavior.

Month	Patient-Days (X)	Maintenance Cost (Y)
Jan	3,000	\$6,200
Feb	4,000	\$6,800
Mar	5,000	\$7,400
Apr	6,000	\$8,100
May	7,000	\$8,900
Jun	8,000	\$9,500

4. Precision Machining Co.'s high-grade titanium alloy is a specialized product where each batch can be uniquely identified. The company's internal management accounting system is evaluating alternative costing methods for strategic decision-making. The transactions for October are detailed below, including unique batch numbers:

Date	Transaction	Batch No.	Units (kg)	Cost per kg (£)	Total Cost (£)
Oct. 1	Beginning Inventory	T-101	400	50.00	20,000
Oct. 5	Purchase	T-215	600	55.00	33,000
Oct. 12	Issue to Job A101		700	?	?
Oct. 18	Purchase	T-308	500	60.00	30,000
Oct. 25	Issue to Job B202		400	?	?
Oct. 29	Purchase	T-419	300	62.00	18,600

Additional Information:

The production manager's reports show that the issue on October 12 for Job A101 specifically used:

- 200 kg from beginning inventory batch T-101
- 500 kg from the October 5 purchase batch T-215

The issue on October 25 for Job B202 specifically used:

- 200 kg from the October 18 purchase batch T-308
- 200 kg from the remaining beginning inventory batch T-101

On October 25, the replacement cost (market price) to purchase more titanium alloy was £61.50 per kg.

Required:

Calculate the total direct material cost charged to Job A101 and Job B202, and the value of the ending inventory on October 31st using the following two methods:

- a) Specific Identification Method
- b) Current Cost (Replacement Cost) Method (Use the replacement cost given for the issue on Oct. 25. For the Oct. 12 issue, assume the replacement cost on that date was £56.50 per kg).

University of Dhaka
Institute of Leather Engineering and Technology
Leather Engineering Department

BSc in Leather Engineering 4th Year 2nd Semester 1st Mid-term Examination – 2025
BUS 402: Cost and Management Accounting

Time: 1 Hour

Marks: 3+3+3+6

Answer all of the following questions:

1. What is Cost Driver? Discuss the cost classification for a manufacturing company with examples.
2. Compare and contrast among Halsey Plan, Rowan Plan, and Taylor's Differential Piece Rate System.
3. Using the information given in the following table, estimate the cost behavior.

Month	Patient-Days (X)	Maintenance Cost (Y)
Jan	3,000	\$6,200
Feb	4,000	\$6,800
Mar	5,000	\$7,400
Apr	6,000	\$8,100
May	7,000	\$8,900
Jun	8,000	\$9,800

4. Precision Machining Co.'s high-grade titanium alloy is a specialized product where each batch can be uniquely identified. The company's internal management accounting system is evaluating alternative costing methods for strategic decision-making. The transactions for October are detailed below, including unique batch numbers:

Date	Transaction	Batch No.	Units (kg)	Cost per kg (£)	Total Cost (£)
Oct. 1	Beginning Inventory	T-101	400	50.00	20,000
Oct. 5	Purchase	T-215	600	55.00	33,000
Oct. 12	Issue to Job A101		700	?	?
Oct. 18	Purchase	T-308	500	60.00	30,000
Oct. 25	Issue to Job B202		400	?	?
Oct. 29	Purchase	T-419	300	62.00	18,600

Additional Information:

The production manager's reports show that the issue on October 12 for Job A101 specifically used:

- 200 kg from beginning inventory batch T-101
- 500 kg from the October 5 purchase batch T-215

The issue on October 25 for Job B202 specifically used:

- 200 kg from the October 18 purchase batch T-308
- 200 kg from the remaining beginning inventory batch T-101

On October 25, the replacement cost (market price) to purchase more titanium alloy was £61.50 per kg.

Required:

Calculate the total direct material cost charged to Job A101 and Job B202, and the value of the ending inventory on October 31st using the following two methods:

- a) Specific Identification Method
- b) Current Cost (Replacement Cost) Method (Use the replacement cost given for the issue on Oct. 25. For the Oct. 12 issue, assume the replacement cost on that date was £56.50 per kg).